Jack (Jiawei) Wang

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Research Motivation

With a strong background in software security analysis and program optimization, my research aims to bridge the gap between static program analysis and real-world software engineering practices. I am particularly interested in how AI-powered tools can augment traditional compiler techniques to detect bugs earlier and optimize software more effectively at scale.

Research Interests

- : Program Analysis and Abstract Interpretation
- : Sparse Representations in Compiler Optimizations
- : AI-Assisted Programming and Intelligent Code Completion
- : Static and Dynamic Analysis for Code Generation Models
- : Security and Reliability in AI-Generated Code
- : Integration of LLMs with Developer Toolchains and IDEs
- : Neurosymbolic Verification: LLM Reasoning Grounded by SMT Solvers

Education

University of New South Wales (UNSW) — **Kensington, NSW**
PhD Candidate in Computer Science — *2023.9–Present*

Syracuse University — **Syracuse, NY**
Master of Computer Science, GPA: 3.9/4.0 — *2016–2018*
Research Assistant: Joined the Spectrum Collaboration Challenge in DARPA

Renmin University of China — **Beijing China**
Bachelor of Science in Computer Science — *2012–2016*

Research Experience

University of New South Wales (UNSW) — **Kensington, NSW**
PhD candidate Researcher — *2023.9–Present*

- Research focus on Abstract Interpretation and Static Analysis.
- Maintained and extended the SVF codebase, adding various advanced features.
- Enhanced front-end support for multiple LLVM versions, including opaque pointer type inference.
- Developed Abstract Interpretation support in SVF for precise static analysis.
- Added new clients for detecting buffer overflow, nullptr dereference, and dangling pointer issues.
- Currently researching:
  - Static analysis support for complex C++ features.
  - Static analysis of recursive functions.
  - Using LLM AI-Agent to assist static analysis.

Selected Research Projects

LLM-Guided Loop Invariant Synthesis — **Python, Z3, vLLM, PyTorch**
UNSW — *2025–Present*

- Co-developed a neurosymbolic framework that decomposes multiphase loops along branch conditions into a search tree, prompts the LLM per branch, composes disjunctive invariants, and localizes Z3 counterexamples to the failing branch for targeted repair; state-of-the-art on 54 multiphase benchmarks and 355/366 linear and nonlinear benchmarks (under review, TSE).
- Built an open-model pipeline on Qwen: SFT and GRPO fine-tuning with a staged verifier reward (init / inductive / post), and multi-GPU vLLM serving (tensor parallelism, FP8 quantization).
- With reasoning (thinking mode) enabled, a locally served open model solved 36/38 dev and 19/29 held-out hard problems (vs. 0/30 for the non-reasoning fine-tuned model), including 15 multiphase programs unsolved by the GPT-5.1-based pipeline.
- Next: SFT on verifier-checked reasoning traces to distill this chain-of-thought capability into smaller models.

Static Verification of LLM Agent Skills **Python, LLM agents**
UNSW *2026–Present*

- Co-developed a neurosymbolic analyzer for agent skills (SKILL.md): recovers a Skill Control Flow Graph from natural-language workflows, uses the LLM to infer per-node post-states, and propagates path-sensitive facts with Hoare-style transfer rules to report violated requirements with counterexample paths.
- On a 202-case benchmark from 20 repositories, improved F1 by 275% over the strongest prompting baseline (Claude Sonnet 4.6) and by over 230% over Claude Code and Codex agents.
- Field study on 14 popular repositories (up to 292K GitHub stars): 94 previously unknown defects reported, 54 fixed or confirmed by maintainers (under review, FSE'27).

Industry Experience

Baidu Inc. **Beijing, China**
Software Engineer *2020–2022*
Paddle Pipeline Serving
Supervisors: Li Zhao, Daxiang Dong

- Designed and implemented DAG-based deep learning model serving, supporting CPU/GPU/Arm/Distributed deployments.
- Integrated with BaiduRPC, GRPC, and PaddlePaddle Inference Lib, optimizing performance across heterogeneous environments.
- Developed Python/Pybind11 wrappers to balance ease of use and performance.
- Patent submission on elastic CTR/PaddleRec (submitted 11/2020).

Baidu Inc. **Beijing, China**
Software Engineer *2019–2020*
Elastic CTR/PaddleRec on Kubernetes
Supervisors: Daxiang Dong, Zhihua Wu

- Designed end-to-end CTR training and inference workflows, implementing distributed CTR training and model pruning.
- Implemented CTR DNN Training in distributed mode, managing 400M+ sparse parameters using PaddlePaddle.
- Developed post-training quantization for embedding tables, achieving 87.5% storage savings at a 3.5% online performance trade-off.
- Built configurable ranking/recall/features modules for CTR models, including User-CF, Item-CF, TDM, Wide&Deep, and DeepFM.
- Developed elastic scheduling mechanisms for dynamically scaling CTR training pods, integrating with MLFlow for visualization.

Publications

Published / Accepted

ISSTA'26: Jiawei Yang, Xiao Cheng, **Jiawei Wang**, Xiapu Luo, Yulei Sui (2026). *Multi-Stage On-Demand Program Slicing for Modular Analysis of Multi-Threaded Programs*. Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026).

OOPSLA'25: **Jiawei Wang**, Xiao Cheng, Yulei Sui (2025). *Towards Efficient Abstract Interpretation by Unleashing the Power of Selective Widening*. Proceedings of the ACM on Programming Languages, Vol. 9, OOPSLA2: 922–949. DOI: 10.1145/3763083.

ICSE'24: Xiao Cheng, **Jiawei Wang**, Yulei Sui (2024). *Precise Sparse Abstract Execution via Cross-Domain Interaction*. Proceedings of ICSE 2024: 109:1-109:12. DOI: 10.1145/3597503.3639220

Under Review

TOSEM: **Jiawei Wang**, Xiao Cheng, Yulei Sui. *Scaling Abstract Interpretation through Interprocedural Sparse Value Flow*. Under review at ACM Transactions on Software Engineering and Methodology (TOSEM). Journal extension of ICSE 2024.

FSE'27: Mingxiu Wang, Ziqi Ding, **Jiawei Wang**, Wei Song, Jingling Xue, Xi Zheng, Xiao Cheng. *When Actions Betray Intentions: Hoare-Inspired Inconsistency Detection in Agent Skills*. Under review at FSE 2027.

TSE: Mingxiu Wang, **Jiawei Wang**, Shuangxiang Kan, Huan Huo, Yuekang Li, Xi Zheng, Xiao Cheng. *Compositional*

Loop Invariant Inference for Multiphase Programs via Structured Neurosymbolic Reasoning. Under review at IEEE Transactions on Software Engineering (TSE).

In Preparation.....

PLDI'27: **Jiawei Wang***, Xiao Cheng*, Yulei Sui. *Beyond Points-To: A Three-Domain Abstract Interpretation for C++ Pointer Lifetimes*. To be submitted to PLDI 2027. (*equal contribution)

ISSTA'27: Qiuyi Feng, **Jiawei Wang**, Yulei Sui. *Dual Approximation of Error Preconditions for Alarm Validation*. To be submitted to ISSTA 2027.

Patents

2018: **[CN201811555963.2]** Convolutional Neural Network Accelerator and Method for Executing Convolution Operation.

2015: **[CN201510895632.3]** A Method of Privacy Enhanced Ticket Authentication Based on 3D Code.

Teaching Experience

University of New South Wales (UNSW) **Kensington, NSW**
Tutor & Marker
Course: CS6131 Software Security Analysis - T2 2024 and T2 2025
Lecturer: Prof. Yulei Sui

- Conducted tutorial sessions and provided academic support to students.
- Designed course assignments based on the SVF (Static Value-Flow) framework for static analysis.
- Developed a CI-based grading system to automate evaluation and feedback.
- Responsible for grading and maintaining consistency across assessments.

University of Technology Sydney (UTS) **Ultimo, NSW**
Tutor & Marker
Course: 41184 Software Analysis Studio – Spring 2024
Lecturer: Prof. Yanjun Zhang

- Led tutorials and guided students through practical software analysis techniques.
- Independently designed assignments and the grading system using the SVF framework.
- Built and managed a CI-based infrastructure for automatic testing and grading of submissions.
- Provided detailed feedback to support student improvement.

References

Prof. Yulei Sui: University of New South Wales
Email: y.sui@unsw.edu.au

Technical Skills

Languages: C++, Python, Rust, Java, Bash

Frameworks & Tools: LLVM, SVF, Clang, Git, Bazel, Docker, Kubernetes, PyBind11

AI/ML Tools: PyTorch, PaddlePaddle, TensorFlow, HuggingFace Transformers

DevOps: CI/CD (GitHub Actions, GitLab CI), MLFlow, Prometheus